

Reduced Physics-Informed Neural Networks for Parametric Poisson Equations

Fast surrogate predictions for repeated one-dimensional Poisson problems

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Projet S6 - P13 · February–June 2026

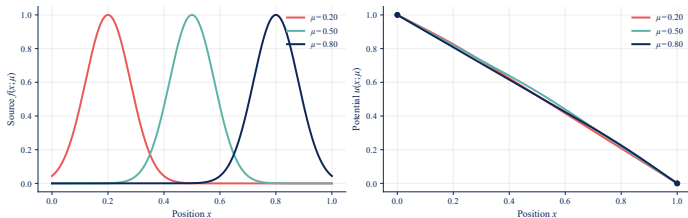
The repeated-query problem

A Poisson equation can describe electric potential between two electrodes.

When the charge distribution changes, the potential changes with it. A parameter study therefore requires many related PDE solves.

Project question

Can previously trained PINNs be reused instead of training a new network for every parameter value?



Source: report_PINN.pdf, Sections 1 and 3; notebooks 03–04. Curves recomputed from the stated Poisson model.

Parametric Poisson equation

$$-\frac{d^2u}{dx^2} = f(x; \mu, \sigma, A) = A \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right], \quad x \in (0, 1)$$
$$u(0) = 1, \quad u(1) = 0$$

Position μ

Moves the localized source through the domain.

Width σ

Controls how sharply the source is concentrated.

Amplitude A

Sets the strength of the source term.

Source: report_PINN.pdf, Eqs. (1)–(6); parameter values in notebooks 06–10.

PINN training uses the differential equation

$$\hat{u}_\theta(x; \lambda) \approx u(x; \lambda), \quad \lambda = (\mu, \sigma, A)$$

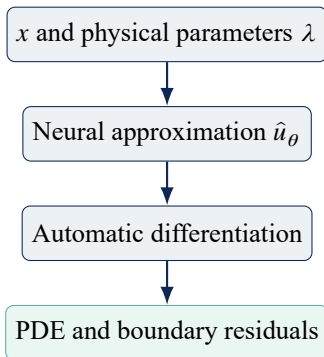
$$r_\theta(x; \lambda) = -\partial_{xx}\hat{u}_\theta(x; \lambda) - f(x; \lambda)$$

$$\mathcal{L}(\theta) = \frac{1}{N_r} \sum_{j=1}^{N_r} |r_\theta(x_j; \lambda)|^2 + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}$$

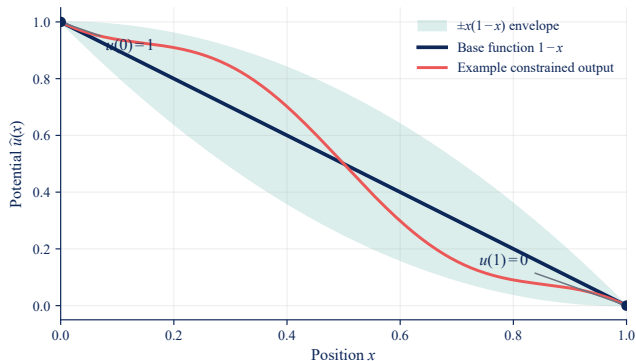
What provides supervision

Automatic differentiation evaluates the PDE residual. Exact solutions remain separate and are used for evaluation.

Source: report_PINN.pdf, Section 3.4; implementations in notebooks 03–05.



Hard constraints enforce both boundary values



$$\hat{u}_\theta(x) = 1 - x + x(1 - x)N_\theta(x)$$

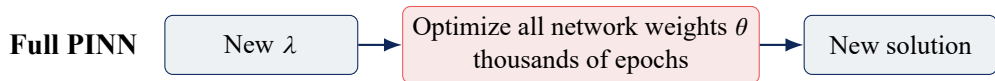
Exact by construction

The correction term vanishes at $x = 0$ and $x = 1$, regardless of the learned network weights.

This removes the boundary penalty and lets training focus on the interior residual.

Source: report_PINN.pdf, Eqs. (12)–(18); hard-constrained implementation in 11discrete.py.

Full PINNs repeat costly optimization



Observed offline cost

One-parameter basis PINNs required approximately 212–237 seconds each in the saved run.

Reduction target

Replace a high-dimensional weight update with a low-dimensional coefficient problem.

Source: Notebook 06, cells 5, 8, and 10. Timings are machine-dependent wall-clock measurements.

Reduced PINN reuses pretrained solution functions

$$\phi_i(x) = \hat{u}_{\theta_i}(x; \lambda_i), \quad u_r(x; \lambda) = \sum_{i=1}^N c_i(\lambda) \phi_i(x)$$

$$\mathcal{L}_{\text{red}} = \mathcal{L}_r + \omega_{\text{BC}} \mathcal{L}_{\text{BC}}$$

$$\mathcal{L}_r = \frac{1}{N_r} \sum_{j=1}^{N_r} \left| - \sum_{i=1}^N c_i \phi_i''(x_j) - f(x_j; \lambda) \right|^2$$

Offline stage

Train full PINNs and store each basis output and second derivative.

Online stage

Optimize only N coefficients and combine the stored basis functions.

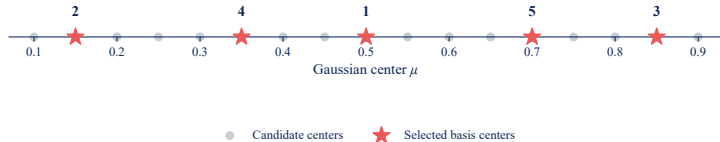
Implementation: direct linear basis combination, not the full GPT-PINN meta-network.

Source: report_PINN.pdf, Eqs. (23)–(30); Chen and Koohy (2024).

Greedy sampling targets the current worst case

$$\lambda_{n+1} = \arg \max_{\lambda \in \Xi_{\text{train}}} \mathcal{L}_{\text{red}}^*(\lambda)$$

The one-parameter run began at the midpoint and added the largest-loss candidate at each step.

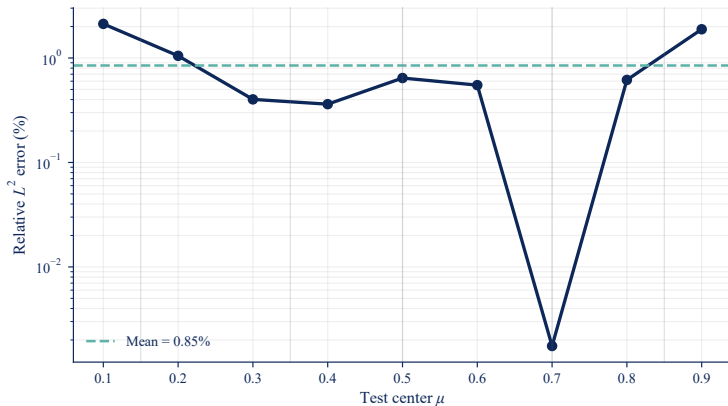


Interpretation boundary

The reduced loss is a practical indicator, not a certified error bound.

Source: Notebook 06, cell 10; report_PINN.pdf, Section 4.4.

Five basis PINNs predict an unseen center accurately



5.585×10^{-3}
relative L^2 error
unseen query $\mu = 0.63$

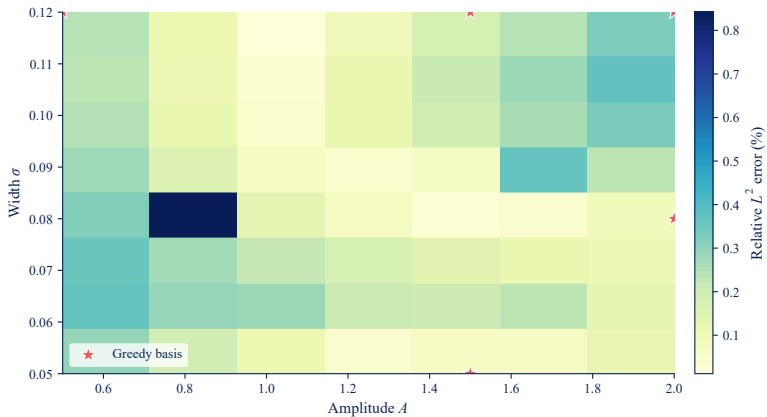
0.85%
mean error
nine-point sweep

2.12%
maximum error
at $\mu = 0.10$

$\varepsilon_{\text{rel}} = \|u_r - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$. The sweep contains interpolation points and basis locations.

Source: Notebook 06, cells 10 and 12. Metrics recomputed from saved notebook outputs.

Two parameters remain accurate on the tested grid



56

test cases

$\mu = 0.50$ fixed

1.91×10^{-3}

mean relative L^2 error

0.19%

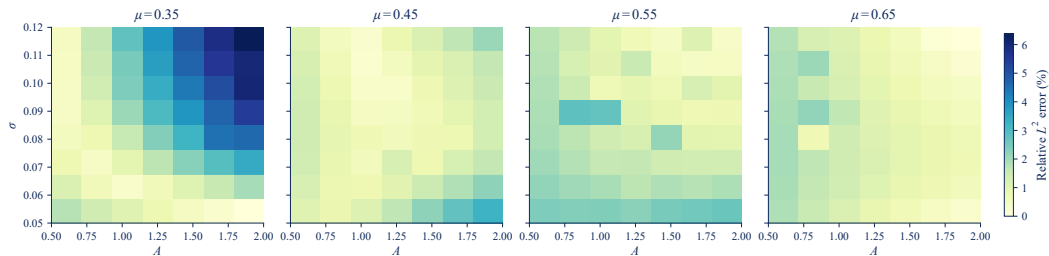
8.419×10^{-3}

maximum relative L^2 error

0.84%

Source: Notebook 07, cell 12. The stars show the five greedy basis parameters.

Three parameters widen the error distribution



224
saved test cases

four μ slices

1.54%
mean error

relative L^2

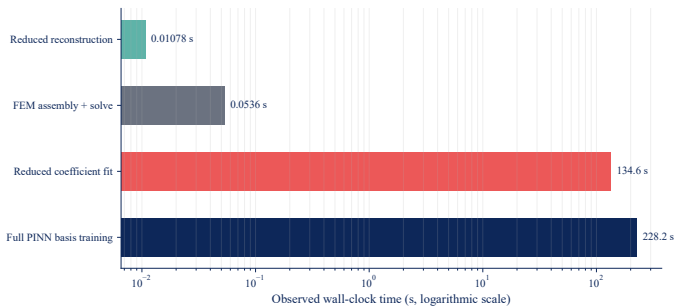
6.36%
maximum error

hardest grid region

A fixed five-function basis must cover a larger solution family as the parameter dimension grows.

Source: Notebook 10, cell 12. Metrics recomputed from all saved (μ, σ, A) outputs.

Reconstruction is fast, but coefficient fitting dominates

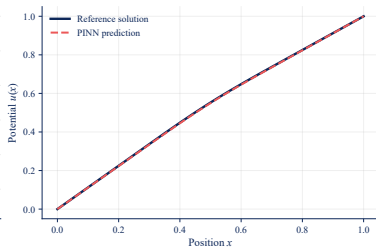
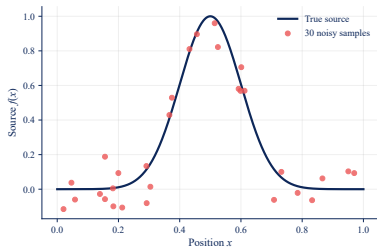


Current implementation does not establish end-to-end speedup over FEM

The reported 0.011 s measures reconstruction after the coefficients are known. The saved coefficient optimization takes 134.600 s, so the full online path is approximately 134.600 s.

Source: Notebook 10, cells 13–14; report_PINN.pdf, Section 6.1. Bars summarize different recorded runs and are not a controlled hardware benchmark.

Noisy discrete source data still yield a smooth solution



30

source measurements

Gaussian noise $\sigma_n = 0.1$

5.69×10^{-3}

final residual loss

5000 Adam epochs

3.51×10^{-3}

relative L^2 error

deterministic rerun

Training minimizes $\mathcal{L}_{\text{data-PDE}} = \frac{1}{30} \sum_{i=1}^{30} |-\hat{u}''_{\theta}(x_i) - f_i|^2$ at the measured source locations.

Source: Reproduced from 11discrete.py with seed 42; report_PINN.pdf, Section 5.

What the project established

Accurate tested surrogate

Five pretrained basis PINNs approximate the tested one- and two-parameter solution families with low relative error.

Useful adaptive construction

Greedy sampling concentrates new basis functions where the current reduced residual remains largest.

Unresolved efficiency claim

Fast reconstruction alone does not make the current online solver faster. Coefficient fitting remains the main bottleneck.

Next research priorities

Replace iterative coefficient fitting with a direct or closed-form reduced solve, validate on unseen parameter regions, add certified error control, and extend the method beyond one-dimensional geometry.

Source: Synthesis of the report, code, and saved experiment outputs.

Implementation settings

Experiment	Network	Full-PINN training		Reduced basis	Test evidence
One parameter μ	3×40 tanh	8000 epochs, points	Adam 400	5 of 17 candidates	9-point sweep plus $\mu = 0.63$
Two parameters (A, σ)	3×40 tanh	8000 epochs, points	Adam 400	5 of 12 candidates	56 saved grid cases
Three parameters (μ, σ, A)	3×40 tanh	8000 epochs, points	Adam 400	5 of 36 candidates	224 saved grid cases
Discrete source	2×40 tanh	5000 epochs	Adam	Not reduced	30 noisy source samples

Source: Notebooks 06, 07, and 10; 11discrete.py. Coefficient fitting uses 1200–4000 Adam iterations per query.

Evidence boundaries

What the experiments support

- Low interpolation error on the saved parameter grids
- Stable hard enforcement of boundary values
- Reuse of a five-function pretrained basis
- Successful recovery from noisy discrete source samples

What remains unproven

- Certified error bounds
- Reliable extrapolation outside the tested domains
- End-to-end speedup over optimized FEM
- Scaling to higher-dimensional PDEs and moving discontinuities

The project uses reduced loss for greedy selection. It does not implement the full GPT-PINN meta-network or a certified reduced-basis estimator.

Source: report_PINN.pdf, Sections 4 and 6; comparison with Reference.pdf.

References



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Source: Reference list in report_PINN.pdf; full GPT-PINN paper in Reference.pdf.